

Block 3 — Chapter 14: Human Factors

Plain-English Study Guide • FAA-H-8083-30 Chapter 14 • prepared for Matthew Johnson

Read this instead of the whole chapter. Every fact is cited to the printed page (e.g. “p. 14-9”) so you can jump back to the book. **Yellow MUST KNOW boxes** are the highest-yield exam items. **Purple boxes** are things the task asked for that are *not actually printed in this -30 chapter* (Aloha 243, IM SAFE) — worth knowing, but confirm with your instructor before trusting them on a chapter-graded worksheet. This is a summary only — no quiz attached.

1. What “Human Factors” Means — and Why It Matters

Human factors = the study of how people interact with their environment, and everything about us (our bodies, our minds, our limits) that affects how well we do a job (p. 14-1). Think of it like this: the airplane is only as safe as the *person* turning the wrench. Human factors is the science of keeping that person from making mistakes.

A **human error** is an **unintentional act of doing a task wrong** that can degrade the system (p. 14-1). The chapter lists three types of human error:

- **Omission** — NOT doing an act or task (you forgot a step).
- **Commission** — doing a task **INCORRECTLY** (you did it wrong).
- **Extraneous** — doing a task you were NOT authorized to do.

And four consequences of human error (p. 14-1): little/no effect → damage to equipment → personal injury → catastrophic.

MUST KNOW — the big number

About **80% of maintenance errors involve human factors** (p. 14-1), and overall **~80% of all aviation accidents are caused by human factors, ~20% by machine/technical failure** (p. 14-29, Fig 14-36). Of all accidents, roughly **75–80% are human error**, and about **12% of those are maintenance related** (p. 14-1). The exam loves these numbers. The whole point of the chapter: *the next big safety gains come from fixing the human, not the technology.*

Why maintenance errors are sneaky: a pilot’s mistake shows up fast and obvious. A mechanic’s mistake is often **latent** — hidden, sitting quietly, sometimes for a long time before it bites (a missed crack, a part installed wrong) — but it can be just as deadly (p. 14-1).

2. A Short History of Human Factors

You don’t need to memorize all of this — but know the milestones the chapter calls out (p. 14-6 to 14-9):

- **~1487 — Leonardo da Vinci.** His *Vitruvian Man* is one of the earliest guides to **anthropometry** (the study of body dimensions). He studied birds to design a flying machine (p. 14-6).
- **Early 1900s — Frank & Lillian Gilbreth** (industrial engineers). Created the **challenge-response / call-back** system (doctor says “scalpel,” nurse repeats “scalpel”). That same speak-it-out-loud protocol is used in aviation today. They also researched **fatigue** (p. 14-6).
- **1903 — Wright Brothers** pioneered human-factors thinking, designing controls a human could actually fly (p. 14-6).
- **WWI & WWII** — focus shifted from *fitting the man to the machine* to *designing the machine for the man* (human capabilities & limitations). The **Hawthorne Effect** showed motivation strongly affects performance (p. 14-7).
- **1978 — Airline Deregulation** brought “do more with less” pressure to maintenance (p. 14-8).
- **1988 — the FAA gets formally involved** with the first Human Factors Issues in Aviation Maintenance conference (p. 14-1). Resource: hf.faa.gov.
- **1990s — “culture change”** at U.S. carriers; 1991 Continental began CRM-type training in maintenance (p. 14-9).

NOT IN THIS CHAPTER — but classic HF history (instructor may mention)

The famous event usually credited with waking up **maintenance** human-factors awareness is **Aloha Airlines Flight 243 (April 28, 1988)** — a Boeing 737 whose fuselage skin tore open in flight from undetected fatigue cracking and corrosion (multiple-site damage). It is the textbook “why maintenance HF matters” case and lines up with the FAA’s 1988 start date above. **This specific event is NOT named in this -30 chapter**, so your AIM worksheet/exam graded against this chapter will not require it — but it’s worth knowing for the FAA written and for context. The chapter’s own real-world cases are Alaska 261, Lufthansa A320, Excalibur 320, China Airlines 611, Colgan Air, and American 1400 (*Fig 14-32, 14-33, 14-37*).

3. The PEAR Model — the “what to worry about” checklist

People • Environment • Actions • Resources

PEAR is a **memory jogger** (mnemonic) for the four big things a maintenance human-factors program should care about (*p. 14-9*). Easiest way to remember it: a **PEAR** is the fruit, and it reminds you of four buckets.

Letter	Stands for	Plain English — what it covers
P	People	The folks doing the work — their physical, physiological, psychological, and psychosocial sides: body size, age, strength, knowledge, training, attitude, fatigue, health, and how they get along with others. Includes planned rest breaks . (<i>p. 14-9, Fig 14-12</i>)
E	Environment	TWO environments: the physical one (heat, humidity, light, noise, the ramp/hangar/shop) AND the organizational one (the company’s culture, communication, supervision, morale, safety culture). (<i>p. 14-9, 14-10, Fig 14-13</i>)
A	Actions	The exact steps a job takes. The tool here is JTA = Job Task Analysis — it identifies the knowledge, skills, and attitudes each task needs. Many authorities require the JTA to be the basis of the company’s general maintenance manual and training plan. (<i>p. 14-10, Fig 14-14</i>)
R	Resources	Anything a tech needs to get the job done — tools, lifts, test equipment, manuals, work cards, parts, training, even a phone or protective clothing. The R element is about spotting the need for more/better resources . (<i>p. 14-10, Fig 14-15</i>)

MUST KNOW

PEAR = People, Environment, Actions, Resources. Environment has two halves (physical + organizational). The Actions tool is the **JTA (Job Task Analysis)**.

4. MEDA — the error-investigation tool

Maintenance Error Decision Aid

MEDA is a **Boeing-developed tool for investigating maintenance errors** (*p. 14-10*). Its whole idea: **find the factors that caused the error and fix them — don’t just blame and re-train the person**. Old-school investigations asked “WHO messed up?” so the real causes never got fixed and the mistake kept coming back. MEDA breaks that “blame and train” cycle.

MEDA rests on three principles (*p. 14-11*):

- **Positive employee intent** — techs want to do a good job and don’t make mistakes on purpose.
- **Contribution of multiple factors** — there’s usually a *chain* of factors, not one.
- **Manageability of errors** — most contributing factors **CAN** be managed.

The five MEDA steps (*p. 14-11*): **1) Event** (pick the error to investigate) → **2) Decision** (was it maintenance related? if yes, continue) → **3) Investigation** (use the MEDA results form) → **4) Prevention strategies** (implement & track fixes) → **5) Feedback** (tell the floor what changed so techs know).

5. The SHEL Model — the four things a person interacts with

Software • Hardware • Environment • Liveware

SHEL puts the **worker (Liveware)** in the **center** and looks at how that person interacts with four things. It was created by Professor **Elwyn Edwards (1972)** and modified by Capt. **Frank Hawkins (1975)** (p. 14-11, 14-12, Fig 14-16). It only looks at interactions that *involve a human*.

Letter	Stands for	Plain English — careful, “Software” is a trap
S	Software (Procedures)	NOT computer programs! Here “software” means the paperwork & procedures — manual layout, checklists, symbols, language, work cards. (p. 14-12)
H	Hardware (Machines)	The physical stuff — location of components, accessibility of parts and tooling. (p. 14-12)
E	Environment (Ambient)	Temperature, humidity, sound, light, and time-of-day. (p. 14-12)
L	Liveware (People)	The technician interacting with other people — managers, peers, family, friends, and self. This is the Liveware-to-Liveware interface : human-to-human. The worker in the center is also Liveware, so SHEL has a Liveware-Liveware link. (p. 14-12)

MUST KNOW — exam trap

In SHEL, **S = Software = procedures/paperwork, NOT computer code. L = Liveware = people.** The **Liveware-to-Liveware interface** is human-to-human interaction (the worker with managers, peers, family, and self). Worker sits in the *center*.

6. The Swiss Cheese Model (Reason’s Model of Accident Causation)

This is **James Reason’s Model of Accident Causation** (introduced 1990, revised 1993) — nicknamed the “**Swiss cheese**” model (p. 14-12).

The analogy: picture several slices of Swiss cheese standing in a row — each slice is a safety defense (a procedure, an inspection, an inspector). Every slice has **holes** (weaknesses). Most of the time a hole in one slice is blocked by solid cheese in the next slice, so nothing gets through. An accident only happens when the **holes in ALL the slices line up** and a hazard passes straight through every defense (p. 14-12).

Two kinds of failures make those holes (p. 14-12, 14-13):

Type	What it is	Example from the chapter
Active failure	An error whose effect is immediate and obvious — the sharp-end event.	An aircraft slips off a lifting jack because the tech placed it wrong (or: a tech falls off a ladder). (p. 14-12, 14-13)
Latent failure	A result of a decision/action made long before ; it can lie dormant a long time before the consequence shows.	Funding wasn’t approved to buy the correct jack, so a wrong jack was in use (or: the ladder was already broken and nobody replaced it). (p. 14-12, 14-13)

MUST KNOW

Swiss cheese = James Reason’s Model of Accident Causation. An accident happens only when the **holes in every slice (defense) line up**. **Active failure = immediate/obvious; latent failure = hidden, decided long before, lies dormant.** An accident is a **chain of events**; break the chain at any link and you stop the accident.

7. Types of Human Error — and Error vs. Violation

The chapter splits errors into **unintentional** vs. **intentional** (p. 14-13):

Term	Plain English
Unintentional error	An accidental slip or wrong judgment — you did NOT mean to. Example: reading a torque value and transposing 26 into 62, or grabbing the wrong work card. It's still a mistake, but not on purpose.
Intentional error = a VIOLATION	When someone knowingly chooses to do something wrong — purposely deviating from safe practices, procedures, standards, or regulations. The chapter says an intentional error “should really be considered a violation.”

MUST KNOW — the difference

An **ERROR** is an **unintentional mistake**. A **VIOLATION** is **intentional** — you knew the rule and broke it anyway. Same wrong outcome, but the difference is **intent**.

Don't confuse those *active/latent* failures from Section 6 with *unintentional/intentional* here — different pairs. Active vs. latent = **when the effect shows**. Unintentional vs. intentional = **whether you meant to**.

8. THE DIRTY DOZEN — the twelve killers (HIGHEST YIELD)

The **Dirty Dozen** = **twelve human factors that most often lead to maintenance errors**. They were identified by **Transport Canada** after a rash of maintenance-related accidents in the late 1980s / early 1990s, then adopted industry-wide (p. 14-13). If you memorize ONE list for the exam, make it this one.

MUST KNOW — the twelve, in order

1 Lack of Communication • 2 Complacency • 3 Lack of Knowledge • 4 Distraction • 5 Lack of Teamwork • 6 Fatigue • 7 Lack of Resources • 8 Pressure • 9 Lack of Assertiveness • 10 Stress • 11 Lack of Awareness • 12 Norms (p. 14-2, 14-13).

The error	What it means (plain words)	The safety net (how to prevent it)	Page
1. Lack of Communication	Information not passed on, or passed on wrong, between techs, shifts, pilots, parts suppliers, or management. Worst at shift change when a half-done job is handed off.	Use logbooks and worksheets to record exactly what is done; do a clear oral AND written turnover at shift change; never assume the work was completed.	p. 14-13
2. Complacency	Over-confidence from doing a task so many times you stop really looking. “I’ve checked this a hundred times, it’s always fine” — so you skip the look.	Always EXPECT to find something wrong; never sign off on anything you did not fully check; always double-check your work.	p. 14-14
3. Lack of Knowledge	Not knowing the system, the current procedure, or that this model differs from one you know. Leads to a faulty repair with catastrophic results.	Only fix what you are trained to fix; make sure the maintenance manual you use is up to date; if you don't know how, ASK someone who does.	p. 14-14, 14-18
4. Distraction	Anything that pulls your mind off the task — a phone call, an aircraft towed by, a family worry. You come back and think you're further along than you are. About 15% of maintenance errors come from distractions.	When you return, go back THREE steps in the procedure to confirm where you left off; use a detailed checklist; never leave tools/ parts lying around — secure them first.	p. 14-14, 14-19
5. Lack of Teamwork	The crew doesn't agree on roles or look out for each other, so things fall through the cracks (a gear swing or ops check needs the whole team working together).	Keep lines of communication open; discuss specific duties when a job needs more than one person; always look out for co-workers with safety in mind.	p. 14-16, 14-20

6. Fatigue	Tiredness from poor sleep, overwork, stress, or working against your circadian rhythm (your body clock). It slows reaction time, memory, and judgment — and you often can't tell you're fatigued. Most dangerous: working alone tired, 1–3 AM.	Be aware of the symptoms in yourself and others; forfeit (hand off) complex tasks when exhausted; eat healthy, exercise, keep a regular sleep pattern. Get 8–9 hrs sleep.	p. 14-16, 14-21
7. Lack of Resources	Missing or low-quality parts, tools, manuals, test equipment, or staff to do the job right. Tempts you to improvise — which can be very costly.	Keep enough parts on hand and order ahead; NEVER substitute an incompatible part to get the job done; keep all equipment properly maintained. (Term: AOG = Aircraft On Ground.)	p. 14-18, 14-23
8. Pressure	The push (from schedule, management, or yourself) to do it faster without mistakes. Self-induced pressure = pressure you put on yourself that wasn't really there.	Make sure the pressure isn't self-induced; speak up if you need more time rather than rushing; ask for extra help when time is tight.	p. 14-22, 14-25
9. Lack of Assertiveness	Not speaking up when something looks wrong. Being assertive = stating your concern positively and productively (NOT aggressively). Staying silent has caused fatal accidents.	Give clear feedback when you see a risk; never compromise your standards; let co-workers voice opinions and accept corrective criticism.	p. 14-22, 14-26
10. Stress	Your subconscious response to demands placed on you. Stressors are physical (heat, noise, bad light, tight spaces), psychological (family, money, conflicts), or physiological (illness, hunger, lack of sleep).	Take time off or a short break when stressed; ask a co-worker to monitor your work; healthy eating, exercise, and enough rest lower stress.	p. 14-24, 14-27
11. Lack of Awareness	Failing to see all the consequences of an action, or what's going on around you ("tunnel vision"), usually from doing the same task so often you stop thinking.	Check whether your work conflicts with an existing mod or repair; always ask co-workers to check your work; even if highly skilled, have someone verify it.	p. 14-26, 14-28
12. Norms	"Norms" is short for "normal" — the unwritten rules of "how we do it here." Unsafe norms (shortcuts, working from memory, skipping paperwork) become accepted as standard and cause accidents.	Make sure everyone follows the SAME standard; remember just because it seems normal doesn't make it correct; the easiest way is often not the approved way.	p. 14-26, 14-29

The chapter calls these safety nets "MITIGATING THE RISK" and shows three for each of the twelve (the green boxes in Figures 14-18 through 14-31).

9. Personal Minimums — the "Am I fit to work?" self-check

A **personal minimums checklist** is a quick honest self-test a tech runs **before** starting work: *am I in good enough shape — body and mind — to do safe work right now?* If the answer is no, you don't start (or you hand the task off). This is the practical, personal side of the Dirty Dozen factors **Fatigue** and **Stress**.

The chapter's own self-care guidance (p. 14-16, 14-17, 14-24, 14-27): watch for fatigue symptoms in yourself; get **8–9 hours of sleep**; avoid complex tasks at the bottom of your circadian rhythm (the **1–3 AM** low); manage stressors; hand off complex work if exhausted; eat well, exercise, limit alcohol/tobacco.

NOT IN THIS -30 CHAPTER — but the classic FAA self-check (worth knowing)

The well-known FAA self-assessment is the **IM SAFE** checklist (from the Pilot's Handbook FAA-H-8083-25 / AC 60-22). It is the standard “personal minimums” acronym. Many instructors teach it alongside maintenance HF even though **this -30 chapter does not print it**:

Letter	Ask yourself...
I	Illness — Am I sick?
M	Medication — Am I on anything that affects me?
S	Stress — Am I overloaded by life/work pressure?
A	Alcohol — Have I been drinking? (none on the job)
F	Fatigue — Am I rested enough?
E	Emotion / Eating — Am I emotionally OK and properly fed?

How a tech uses it: run IM SAFE in your head before clocking onto a job. Any “yes” to Illness/Medication/Stress/Alcohol/Fatigue (or “no” to being emotionally steady and fed) is a flag to slow down, get help, or step away — because a maintenance mistake is latent and may not show up until someone is flying. **Confirm with your AIM instructor whether IM SAFE is on your graded material for this chapter — it is FAA-standard but not in this -30 text.**

10. The 5 Things to Remember + Quick Acronym Card

1. **~80% of maintenance errors are human factors** — not lack of skill. Fixing the human is the next big safety win.
2. **The Dirty Dozen** are the 12 most common human-factor traps (Transport Canada). Know them cold, plus a safety net for each.
3. **PEAR** = People, Environment, Actions, Resources — the four buckets to manage.
4. **SHEL** = Software (procedures, NOT code), Hardware, Environment, Liveware (people) — worker in the center.
5. **Swiss cheese** = Reason's model: accidents happen when the holes in every defense line up. Active error = obvious now; latent = hidden from a past decision.

Acronym / Model	Spelled out	One-line job
PEAR	People • Environment • Actions • Resources	Checklist of the 4 things an HF program manages (14-9)
SHEL	Software • Hardware • Environment • Liveware	The 4 things the worker (center) interacts with (14-11)
MEDA	Maintenance Error Decision Aid	Boeing tool to investigate errors, not blame people (14-10)
JTA	Job Task Analysis	Lists knowledge/skills/attitudes a task needs (the “A” in PEAR) (14-10)
AOG	Aircraft On Ground	A grounding problem; rush to get parts to return to service (14-19)
Dirty Dozen	12 human factors (see Section 8)	The 12 most common causes of maintenance error (14-13)
Swiss Cheese	Reason's Model of Accident Causation	Holes in defenses must line up for an accident (14-12)
IM SAFE	Illness • Medication • Stress • Alcohol • Fatigue • Emotion	Personal-minimums self-check (FAA-H-8083-25, NOT in this -30 ch.)